

"SOLANGE": Quantum Computing and AI Against Non-Druggable Mutations - A New Frontier in NSCLC Oncology

Scientific Oncology Legacy Advancing Undruggable Ground-state Energetics

BLAIS Challenge Award Proposal, August 2026

Budget Template for BLAIS Challenge

Object Code		Amount
SO508	Other Current Expenses: running simulations on the IBM Heron R3 quantum platform	\$15,000
SO179	Graduate Student Wages	\$9,523.81
SO540	Fringe Benefits: Student = 5%	\$476.19
	TOTAL:	\$25,000.00

Description:

SO508 - Running simulations on the IBM Heron R3 quantum platform

Billing rate: \$1.60/second QPU execution time.

Calibration (measured): warm-up circuits at 4-8 qubits ran 21s–125s, giving a per-qubit rate of 5.25-15.6 s/qubit (real range from the two calibration endpoints, not an assumed constant).

Applied to R175H's full target scale (108 qubits - the largest AVAS-tested criterion; classification currently provisional Class A pending final DMRG confirmation): per-circuit execution time is estimated at 567-1,685 seconds (\$907–\$2,696/circuit) depending on how the calibration scaling holds at this larger size.

Requested budget: \$15,000, funding approximately 6-17 sqDRIFT sampling circuits at R175H's full target scale.

This is framed explicitly as a first exploratory tranche, not a convergence guarantee: unlike C275F, where DMRG independently confirms the entanglement stays classically tractable so far, R175H's classification as quantum-necessary means no classical method can tell us in advance how many samples convergence requires - that is precisely what makes it Class A.

Running out of this budget before convergence is itself a valid, informative result, consistent with how this pipeline already treats classical DMRG runs cut short by a time budget (PROVISIONAL, not a failure).

Note on C275F: its Class B designation is measured, not assumed - but only within the active-space range tested so far (up to 48 electrons / 56 qubits, a 5Å radius around the mutated residue, no Zn^{2+} -coordination shell, no chemist-defined criterion yet). That range is not the full binding site, so it is not yet known whether C275F stays Class B once a larger or chemist-defined active space is measured. This means the current single Class-A candidate (R175H) may not remain the only one requiring hardware time - a second request may follow once C275F's fuller active space is tested.

SO179 - Graduate Student Wages

Graduate Student will design, plan, code and execute the simulations on IBM quantum computer. Following the results, a detailed written reports will be executed as part of the project.